Sequences — problem 9.11

A. Engel — Problem-Solving Strategies · 1998

$a_1 = a_2 = 1$, $a_n = 1/a_{n-1} + 1/a_{n-2}$, $n \geq 2$. Find the $\lim_{n\to\infty} a_n$ and the convergence rate.